Upasana Puranik

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Education

University of California, Berkeley Expected May 2028

Bachelor of Science in Electrical Engineering and Computer Science (EECS) *Berkeley, CA*

- **Relevant Coursework:** CS 170 (Algorithms & Complexity), CS 161 (Computer Security), CS 70 (Discrete Math & Probability), EECS 127 (Optimization), CS 61A/B (Programming & Data Structures), CS 61C (Computer Architecture), EECS 16A/16B (Signals & Circuits), CS 194 (AI and ML applications)), ECON C110 (Game Theory), IEOR 172 (Probability & Risk)

Experience

Berkeley AI Research (BAIR) August 2025 – December 2025

Computational Audio Researcher *Berkeley, CA*

- Designed and built a technique-aware audio-to-tablature transcription system (TART) that captures expressive guitar gestures (slides, bends, hammer-ons) and outputs playable tablature.
- Designed, trained, and evaluated deep learning models across audio-to-MIDI conversion, technique detection, and transformer-based fretboard mapping, using large curated datasets and spectrogram analysis.
- Improved transcription accuracy (F1, onset timing) by 40% and reduced unplayable chord predictions through quantitative error analysis; work submitted to IEEE conference and published on arXiv.

Flexify.ai June 2025 – Aug 2025

AI/ML Research Engineer Intern *Berkeley, CA*

- Built and deployed custom multi-AI agents using online reinforcement learning for supply chain sourcing and planning.
- Created and maintained high-quality training datasets for supervised fine-tuning and online learning pipelines, improving with tools like LangChain model accuracy and reliability.

UC Santa Cruz June 2023 – Aug 2023

Research Intern *Santa Cruz, CA*

- Developed novel data generation pipeline to produce gold standard that trained Q&A Bots with BERT.
- Analyzed datasets and researched deep learning models with Python - findings used in a Ph.D. research thesis.

SafeStride May 2025 – Present

CEO & Co-founder *Berkeley, CA*

- Spearheading the development of a real-time safety companion app that connects students for safe walks/commutes 24/7.
- Assembled and led a cross-functional team (engineering + business) to create safer campuses for all students.

Projects

Ngordnet | *Java*

- Created a linguistic analysis tool using Google N-Grams + WordNet with custom graph structures for queries.

Woosh (Berkeley SkyDeck Startup) | *PostgreSQL, Supabase, REST APIs, Git*

- Built backend services, implemented secure authentication & user data storage integrated with a React frontend.

Scheme Interpreter | *Scheme, Git*

- Built a Python-based Scheme interpreter supporting math, special forms, and recursion with tail call optimization.

SomniWatch | *Arduino, Swift, Edge Impulse, ML*

- Created a wearable drowsy driving detection device using biosensors with an embedded ML model and app.

Technical Skills

Languages: Python, Java, C, RISC-V Assembly, SQL, JavaScript, HTML, CSS, Scheme,
ML Frameworks: PyTorch, TensorFlow, Edge Impulse
Systems & Tools: Git, Linux, Arduino, LTSpice, Docker, Supabase, JypterNotebook
Awards: 3x AIME Qualifier, USACO Silver, placed 1/900 at Synaptic hack-a-thon, Congressional Award Silver Medal, National Science Bee Qualifier, 2x Gold Presidential Award, First Class Honors for Royal Consevatory of Music
Other: Valkyrie Robotics Team, ASDRP Machine Learning Intern, Eagle Scout & Den Chief (founder of largest girls' BSA troop), Author & DP Publisher's founder (books avaliable in 8500+ libraries worldwide)